

UI

Tailwind

Customization

Tailwind is meant to be customized.

You should:

- Define brand colors
- Define spacing scale
- Define breakpoints
- Define container sizes
- Define font stack

If you don't customize it...

You're not using Tailwind properly.

Utility-First ≠ Messy Code

Bad junior usage:

```
<div class="flex justify-center items-center bg-blue-500 text-white px-6 py-3 rounded-lg shadow-lg hover:bg-blue-600 transition duration-200">
```

Senior usage:

```
<button class="btn-primary">
```

With:

```
@layer components {  
  .btn-primary {  
    @apply flex justify-center items-center bg-primary text-white px-6 py-3 rounded-lg shadow-lg hover:bg-primary-dark transition;  
  }  
}
```

👉 Use @layer components for:

- Reusable UI patterns

- Buttons
- Cards
- Inputs

Utility-first doesn't mean no abstraction.

Variants Are a System

Tailwind variants:

- hover:
- focus:
- group-hover:
- peer:
- dark:
- md:
- lg:

Senior-level thinking:

You are building state-driven styling.

Example:

```
<div class="group">  
<p class="group-hover:text-red-500">
```

Custom Colors / Spacing / Tokens

These go in:

 `tailwind.config.js`

This is where your design system lives.

Example:

```
// tailwind.config.js

export default {
  content: ["/index.html", "/src/**/*.{js,ts,jsx,tsx}"],
  theme: {
    extend: {
      colors: {
        primary: "#2563eb",
        "primary-dark": "#1e40af",
        brand: {
          50: "#eff6ff",
          500: "#3b82f6",
          900: "#1e3a8a",
        },
      },
      spacing: {
        13: "13px",
      },
      fontFamily: {
        sans: ["Inter", "sans-serif"],
      },
    },
  },
  plugins: [],
};
```

2 Custom Components with @layer

These go in:

 src/index.css (or src/globals.css)

This file must include:

@tailwind base;

@tailwind components;

@tailwind utilities;

Then below that:

```
@layer components {  
  .btn-primary {  
    @apply bg-primary text-white px-6 py-3 rounded-lg  
    hover:bg-primary-dark transition;  
  }  
  
  .card {  
    @apply bg-white rounded-xl shadow-md p-6;  
  }  
}
```

Framer Motion

Framer Motion Cheat Sheet (Practical Version)

1 Basic Setup

`npm install framer-motion`

`import { motion } from "framer-motion"`

Replace any HTML element:

`<motion.div />`

`<motion.button />`

`<motion.li />`

2 Basic Animation (Most Important)

```
<motion.div
  initial={{ opacity: 0 }}
  animate={{ opacity: 1 }}
  transition={{ duration: 0.5 }}
/>
```

◆ Meaning

Prop	What it does
initial	Starting state
animate	Final state
transition	How it moves

3 Common Properties You Animate

opacity

x

y
scale
rotate

Example:

```
initial={{ opacity: 0, y: 50 }}  
animate={{ opacity: 1, y: 0 }}
```

4 Hover & Tap (Super Useful)

```
<motion.button  
  whileHover={{ scale: 1.1 }}  
  whileTap={{ scale: 0.9 }}  
>  
  Click  
</motion.button>
```

No state needed.

5 Exit Animation (Important)

```
import { AnimatePresence } from "framer-motion"  
  
<AnimatePresence>  
  {isOpen && (  
    <motion.div  
      exit={{ opacity: 0 }}  
    />  
  )}  
</AnimatePresence>
```

Without AnimatePresence, exit won't work.

6 Layout Animation (Smooth Reordering)

```
<motion.div layout />
```

Used for:

- Lists
- Expanding cards
- Accordions

Makes position changes animate automatically.

7 Stagger Children

```
const container = {  
  visible: {  
    transition: {  
      staggerChildren: 0.2  
    }  
  }  
}
```

Parent controls timing.

Children animate one by one.

8 Continuous Animation (Loop)

```
<motion.div  
  animate={{ scale: [1, 1.2, 1] }}  
  transition={{ repeat: Infinity, duration: 1 }}  
>
```

9 Spring Animation (Feels Better)

```
transition={{  
  type: "spring",  
  stiffness: 200,  
  damping: 15  
}}
```

- Higher stiffness → faster
- Lower damping → more bounce

10 Performance Rule (Very Important)

Animate:

- opacity
- x / y
- scale
- rotate

Avoid animating:

- width
- height
- top
- left

(They cause layout recalculation.)

Mental Model

Framer Motion works in layers:

1. initial / animate → on mount
2. whileHover / whileTap → interaction
3. exit → unmount
4. layout → position changes

That's 90% of what you need.